

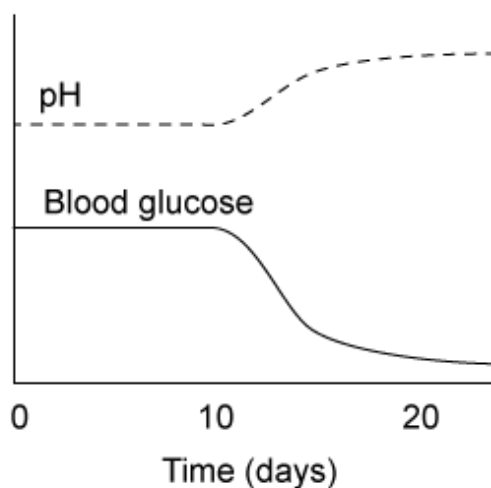
The liver plays a central role in maintaining blood glucose homeostasis. It contains the highest concentration of glycogen of any organ in the body and is one of the few organs that can regenerate glucose from metabolic byproducts. The liver can replenish glucose in muscle and brain tissue during times of fasting, and it can help prevent lactate accumulation during intense physical activity.

Von Gierke disease is a rare autosomal recessive disorder that arises from inactivating mutations in the liver enzyme glucose 6-phosphatase (G6Pase). It affects roughly 1 in 100,000 individuals, and the resulting loss of enzymatic activity leads to hypoglycemia (low blood sugar) that cannot be counteracted by catabolic hormones like glucagon. Von Gierke can also cause an increase in blood acidity concurrent with lactate build-up. Patients typically become symptomatic shortly after birth with convulsions, hyperventilation, and tremors. Left untreated, patients may develop gout, osteoporosis, and life-threatening complications such as kidney failure and liver tumors.

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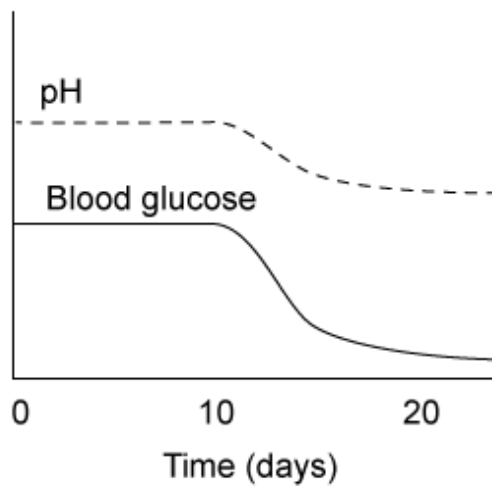
Which of the following graphs best reflects the expected average blood glucose and pH levels over time for group B mice?

☐ A.



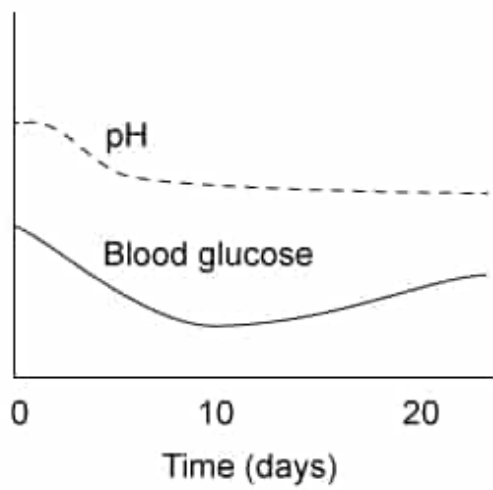
[14%]

✓ ☒ B.



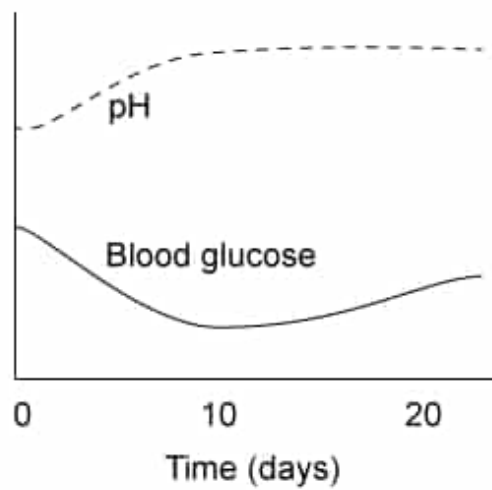
[76%]

☐ C.



[4%]

☐ D.

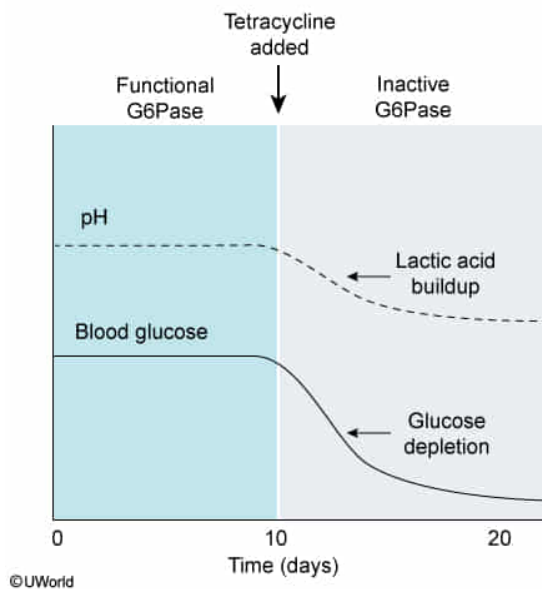


[4%]

Correct

76% answered correctly

Explanation:



Blood metabolite and pH levels are normally tightly controlled. As the passage states, however, **glucose 6-phosphatase (G6Pase) inactivation** leads to **decreased blood sugar** and **increased blood acidity**. Group B mice were given tetracycline, which inactivated G6Pase 10 days into the experiment. Therefore, group B mice should maintain nearly constant glucose and pH levels prior to day 10 and should demonstrate decreased blood glucose and increased acidity after day 10. Because increased acidity means a **decrease in pH**, the correct graph should show a decrease in both glucose and pH levels, starting at day 10.

(Choice A) The blood acidity should increase, meaning that the pH should decrease, starting at day 10.

(Choices C and D) Glucose and pH levels should remain essentially constant prior to administration of tetracycline at day 10.

Educational objective:

Blood metabolite and pH levels are tightly controlled under normal conditions but can be altered by inactivation of metabolic enzymes such as glucose 6-phosphatase. An increase in acidity corresponds to a decrease in pH.

Time spent: 0 sec

Subject:
Biochemistry

QID: 400680

System:
Metabolic Reactions

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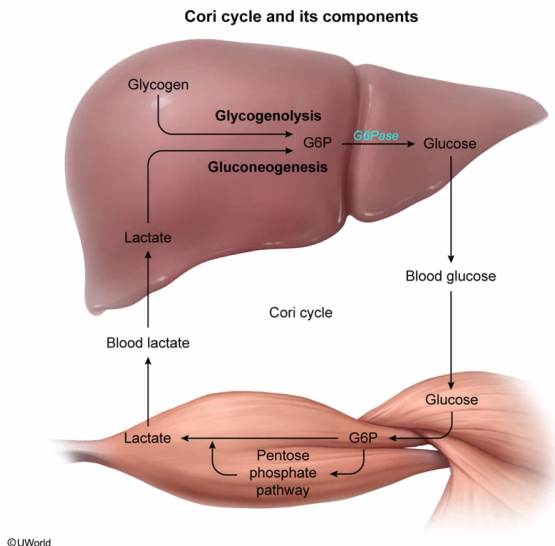
Which of the following processes would be directly impaired in von Gierke disease?

- I. Glycogenolysis
 - II. Gluconeogenesis
 - III. Pentose phosphate pathway
- ✓ ☒ A. I and II only [51%]
☐ B. I and III only [8%]
☐ C. II and III only [23%]
☐ D. I, II, and III [16%]

Correct

50% answered correctly

Explanation:



Maintaining blood glucose levels is critical for proper body function, so **several biological processes** exist to **maintain glucose levels** between meals. They take place predominantly in the liver, and **all produce glucose 6-phosphate (G6P)**. G6P must be dephosphorylated by G6Pase to be released as glucose into the bloodstream, so all these processes will be directly impaired by the inactivation of G6Pase. They include the following:

- **Glycogenolysis** is the **breakdown of glycogen** into glucose. It is the first source of glucose during fasting and requires G6Pase in its final step (**Number I**).
- **Gluconeogenesis** is the **synthesis of glucose** from pyruvate and other precursors. It is essentially glycolysis in reverse with a few differences, including the conversion of pyruvate to oxaloacetate and the use of **phosphatases** such as G6Pase instead of kinases to remove phosphate groups (**Number II**).
Gluconeogenesis and glycolysis are connected by **the Cori cycle** in which lactate, produced in the muscle during anaerobic glycolysis, is sent to the liver. There it is converted to pyruvate and undergoes gluconeogenesis. The newly synthesized glucose is then returned to the muscles through the bloodstream.

(Number III) The **pentose phosphate pathway** is required for nucleotide and NADPH synthesis. Although the pathway includes G6P as a substrate, it does not require G6Pase activity, so it will not be directly impaired by von Gierke disease.

Educational objective:

Several biological processes, including gluconeogenesis, glycogenolysis, and the Cori cycle, help buffer blood glucose levels. All these processes require glucose 6-phosphatase (G6Pase) to catalyze the final step to release free glucose. The pentose phosphate pathway uses glucose 6-phosphate as a substrate, but it does not produce free glucose or require dephosphorylation and is unaffected by changes in G6Pase activity.

Subject:
Biochemistry

System:
Metabolic Reactions

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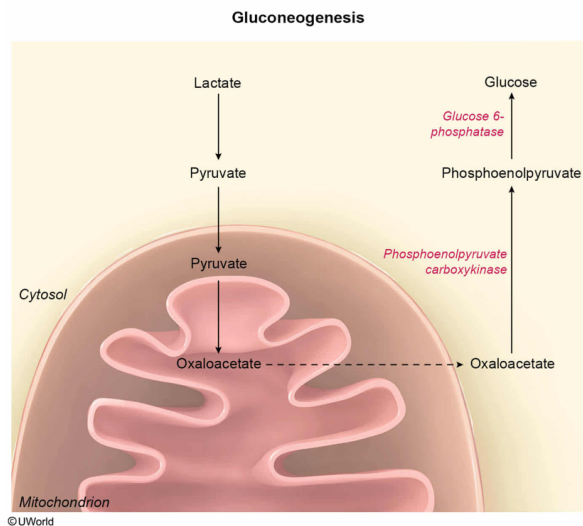
Inactivation of which enzyme is most likely to result in physiological effects similar to those of von Gierke disease?

- ☐ A. Glucose 6-phosphate dehydrogenase [19%]
- ☐ B. Pyruvate decarboxylase [19%]
- ✓ ☒ C. Phosphoenolpyruvate carboxykinase [56%]
- ☐ D. Acyl-CoA dehydrogenase [4%]

Correct

56% answered correctly

Explanation:



The physiological effects of von Gierke disease include low blood glucose and high lactate levels as G6Pase inactivation shuts down the pathways of their release and uptake by the liver. One of these pathways, gluconeogenesis, consumes lactate to produce glucose, and **blocking any enzyme of gluconeogenesis** would therefore likely lead to effects **similar to those of von Gierke** disease. Phosphoenolpyruvate carboxykinase (PEPCK) catalyzes the second step in gluconeogenesis, the conversion of oxaloacetate to phosphoenolpyruvate, and its inactivation can indeed lead to both lactate buildup and glucose depletion in the blood.

(Choice A) Glucose 6-phosphate dehydrogenase (G6PDH) is an enzyme of the pentose phosphate pathway. Although G6PDH uses glucose 6-phosphate as a substrate, it is not involved in gluconeogenesis.

(Choice B) Pyruvate decarboxylase converts pyruvate to acetaldehyde as the first step in the production of ethanol by yeast. Humans do not have this enzyme.

(Choice D) Acyl-CoA dehydrogenase catalyzes the first step in **fatty acid catabolism** and does not directly affect the ability of the liver to produce glucose or consume lactate.

Educational objective:

Inactivating one enzyme in a metabolic pathway often results in physiological consequences similar to those of inactivating another enzyme in the same pathway. Products downstream of the inactivated enzyme will not be produced, and upstream molecules will accumulate.

Time spent: 0 sec
Subject:
 Biochemistry

QID: 400682
System:
 Metabolic Reactions

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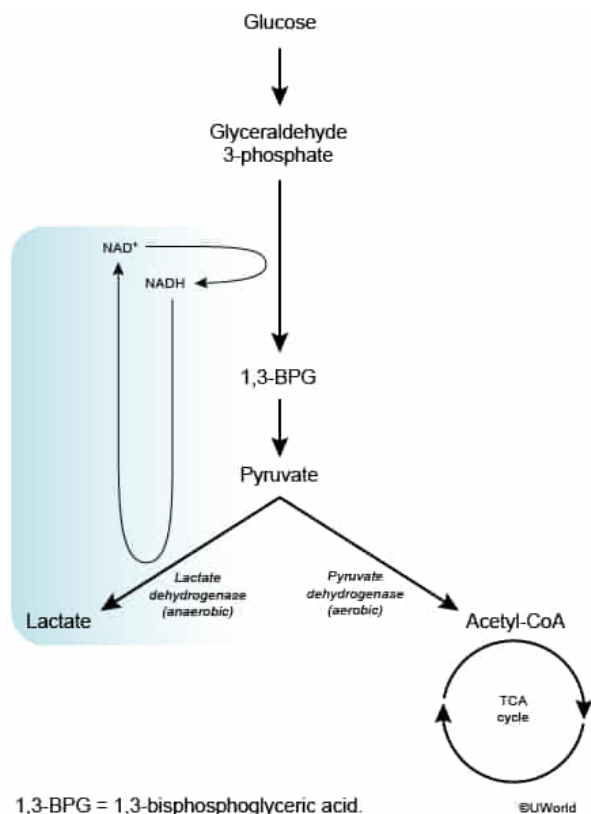
Which cofactor is regenerated by lactate synthesis?

- ✓ ☒ A. NAD⁺ [71%]
☐ B. NADH [18%]
☐ C. NADP⁺ [4%]
☐ D. NADPH [5%]

Correct

72% answered correctly

Explanation:



During glycolysis, NAD^+ is reduced to NADH as glyceraldehyde 3-phosphate is oxidized to 1,3-bisphosphoglycerate; therefore, **NAD^+ is required for glycolysis** to continue. Under aerobic conditions, NAD^+ can be regenerated by oxidation of NADH in the electron transport chain (ETC). Under **anaerobic conditions**, the ETC cannot regenerate enough NAD^+ , and another source is required. Under these conditions, **lactate synthesis** takes place and oxidizes NADH to NAD^+ as pyruvate (the final product of glycolysis) is reduced to lactate.

(Choice B) Lactate synthesis *consumes* NADH to regenerate NAD^+ .

(Choices C and D) NADP^+ and NADPH are used in the pentose phosphate pathway and anabolic pathways such as fatty acid synthesis, but not in glycolysis. Therefore, they do not need to be regenerated or consumed by lactate synthesis.

Educational objective:

Lactate synthesis is coupled to the synthesis of NAD^+ from NADH. Under anaerobic or oxygen-poor conditions, this process provides the NAD^+ necessary to continue glycolysis.

Time spent: 0 sec
Subject:
Biochemistry

QID: 400683
System:
Metabolic Reactions

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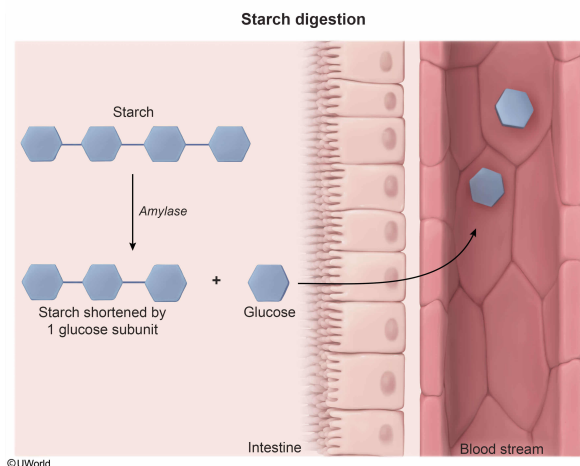
Treatment with which of the following could help counteract hypoglycemia in patients with von Gierke disease?

- ☐ A. Insulin [9%]
- ☐ B. Vitamin A [3%]
- ☐ C. Epinephrine [41%]
- ✓ ☒ D. Starch [45%]

Correct

46% answered correctly

Explanation:



Hypoglycemia can normally be countered with **hormones** that upregulate gluconeogenesis and glycogenolysis. Normally, **glucagon** would help counteract hypoglycemia by upregulating gluconeogenesis and glycogenolysis in the liver. However, the passage states that von Gierke disease is **insensitive to hormones** such as glucagon. This is because the glucose 6-phosphate produced during gluconeogenesis and glycogenolysis will not be hydrolyzed to free glucose, regardless of hormone levels. Blood glucose levels must therefore be **controlled by diet** instead.

Starch is a polysaccharide composed of **several linked glucose molecules**. It is slowly broken down to its constituent glucose molecules in the digestive tract by the enzyme amylase. As glucose molecules are released from starch, they enter the bloodstream and help counteract hypoglycemia.

(Choice A) **Insulin** induces the uptake of glucose from the blood into cells and upregulates glucose storage (glycogen production) rather than synthesis (glycogenolysis). It would increase the severity of hypoglycemia.

(Choice B) Vitamin A is a lipid-soluble vitamin that is required for vision, immune system maintenance, and growth. It is not involved in glucose homeostasis.

(Choice C) Like glucagon, the hormone epinephrine increases blood glucose levels in individuals *without von Gierke disease*. However, the passage states that von Gierke disease renders patients insensitive to hormones. Because gluconeogenesis and glycogenolysis cannot be completed due to impaired G6Pase activity, epinephrine will likely *exacerbate* the patient's symptoms as more G6P builds up.

Educational objective:

Genetic mutations that inhibit gluconeogenesis and glycogenolysis can prevent effective hormonal control of blood glucose levels. Dietary supplementation with a glucose source such as starch can help maintain glucose levels in these cases.

Time spent: 0 sec
Subject:
Biochemistry

QID: 400684
System:
Metabolic Reactions

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Group A mice, but not group B mice, could use which of the following dietary supplements to synthesize glucose and help maintain blood glucose levels after day 10?

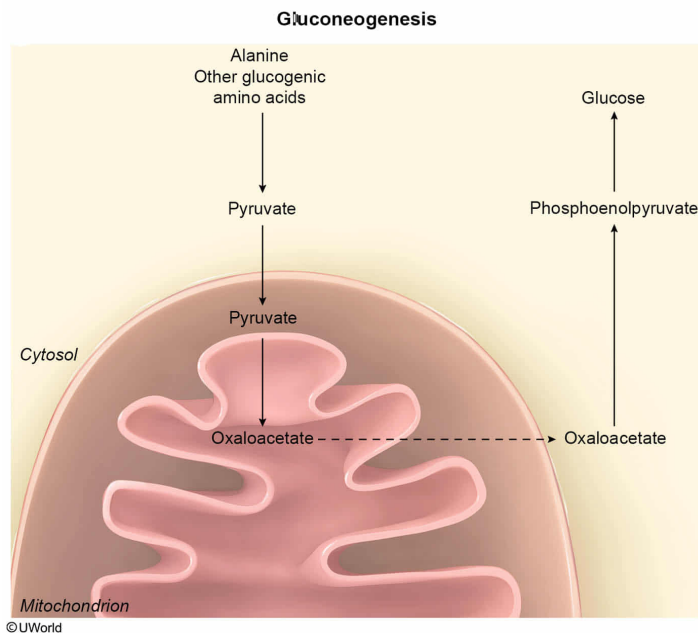
- ☐ A. Cholesterol [5%]
- ✓ ☒ B. Alanine [41%]

- ☐ C. Acetyl-CoA [35%]
- ☐ D. Palmitic acid [17%]

Correct

42% answered correctly

Explanation:



The question asks which treatment can help synthesize and maintain blood glucose levels in group A mice, but *not* in group B mice.

Therefore, a supplement that can be used to produce glucose in group A mice, but whose pathway is impaired in group B mice, is needed.

Mice in group A were not given tetracycline at any point and so **have functional G6Pase**. They therefore will be able to use any metabolites that can enter **gluconeogenesis** to help synthesize glucose to sustain their blood glucose levels. **Amino acids** (except for leucine and lysine) can be **converted into glucose** through gluconeogenesis. For example, alanine can be converted to pyruvate by **deamination**, and pyruvate can then enter gluconeogenesis. Mice in group A will be able to use alanine to make glucose in the liver, thereby maintaining blood glucose levels.

Mice in group B were given tetracycline at day 10 and will be **unable to carry out gluconeogenesis** after that point because the glucose-6-phosphate will not be able to be hydrolyzed to free glucose. Therefore,

glucogenic substrates such as alanine will not help maintain their blood glucose levels.

(Choice A) Cholesterol is a lipid used as a structural molecule in cell membranes and in the synthesis of steroid hormones. It is not used in glucose synthesis.

(Choice C) Acetyl-CoA *cannot* provide additional carbon for glucose synthesis. Although acetyl-CoA can be converted to oxaloacetate through the citric acid cycle, it must first **combine with an oxaloacetate molecule** to do so, thereby consuming it; the two carbons gained by adding acetyl-CoA to oxaloacetate are effectively lost as CO₂ during the cycle, so no net oxaloacetate is produced.

(Choice D) Most fatty acids, including palmitic acid, are **broken down to acetyl-CoA**, which cannot provide additional carbons for glucose synthesis.

Educational objective:

Blood glucose levels can be buffered by gluconeogenesis. Most amino acids are glucogenic, with the exception of leucine and lysine.

Individuals who cannot carry out gluconeogenesis cannot buffer blood glucose with these metabolites.

Time spent:0 sec

Subject:
Biochemistry

QID:400685

System:
Metabolic Reactions